A Cost–Quality Trade-off Study of Lightweight Generative Adversarial Networks for 64 × 64 Hematological Synthesis

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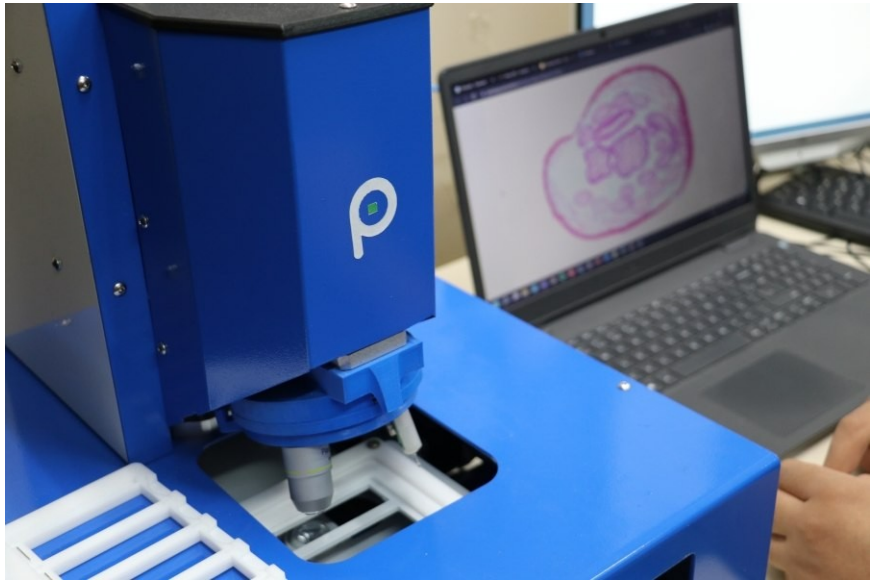
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The problem: a real-world use case



Clinical care reaching a remote community.
U.S. Navy photo, public domain (Wikimedia Commons).



Automated microscope in routine laboratory use.
PickCells, at Hospital do Câncer de Pernambuco.

Blood analysis has to reach collection points far from any central laboratory.

State-of-the-art image models assume hardware those places will never have.

How light can a generative model be, and still support diagnosis?

Motivation and deployment constraints

Hematology pipelines need synthetic cells for three concrete jobs [7], [8]:

- augmentation for classifiers facing class imbalance [9]; the smallest BloodMNIST class has 849 images [14]
- GAN-based anomaly detection, where the generator cost is paid on every sample
- rapid prototyping on the consumer GPUs typical of academic labs

In all three, the binding constraint is a latency and VRAM envelope, not accuracy alone

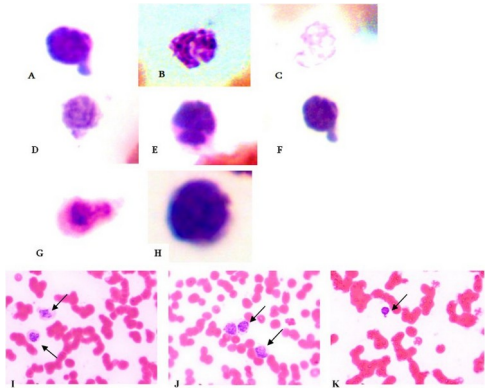
All experiments run on a single consumer GPU: NVIDIA RTX 3050, 8 GB

about US$ 150 to 180

entry-level street price,
not a data-centre GPU



First look at the sample under the microscope.
Photo: N. Stojkovic, CC BY 2.0 (Wikimedia Commons).



Lymphocyte anomalies, COVID-19 blood films.
Karapetyan et al., Heliyon 2024, CC BY 4.0.

[7] Yi et al., Medical Image Analysis 2019
[8] Skandarani et al., Journal of Imaging 2023
[9] Frid-Adar et al., IEEE ISBI 2018
[14] Yang et al., Scientific Data 2023 (MedMNIST v2)

Related work

two largely separate ecosystems in medical imaging: clinical-grade fidelity vs. low-end deployment envelopes

The GAN lineage

- Goodfellow 2014 [1], then DCGAN [2], [3], then the style-based families [4]
- StyleGAN2-Ada [5]: the canonical high-fidelity reference
- Lightweight branch: FastGAN [6], the most-cited instance

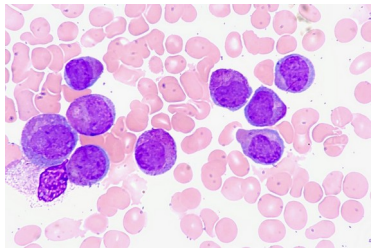
Hematology-specific work

- SyntheticCellGAN [10] and MoE-cGAN [11]: high quality at moderate compute
- Eckardt et al. [12]: StyleGAN2-Ada on bone-marrow smears
- CytoDiffusion [13]: diffusion foundation model, 10-rater evaluation

The under-served niche

- BloodMNIST [14], [15] is the benchmark of record at 64 × 64
- FastGAN was designed for im_size ≥ 256 [6], with no public 64 × 64 implementation
- Custom lightweight GANs are compared against StyleGAN2-Ada two at a time [10], [16]

The multi-objective comparison machinery exists [17], but is rarely applied to medical-imaging GAN benchmarks



Bone-marrow aspirate smear (illustrative).
Ed Uthman, CC BY 2.0, Wikimedia Commons.

[1] Goodfellow et al., NeurIPS 2014
[2] Radford et al., ICLR 2016 (DCGAN)
[3] Salimans et al., NeurIPS 2016
[4] Karras et al., CVPR 2020 (StyleGAN2)
[5] Karras et al., NeurIPS 2020 (StyleGAN2-Ada)
[6] Liu et al., ICLR 2021 (FastGAN)
[10] Barrera et al., CMPB 2023 (SyntheticCellGAN)
[11] Kwiek et al., Electronics 2025 (MoE-cGAN)
[12] Eckardt et al., Leukemia 2022
[13] Deltadahl et al., Nature Machine Intel. 2025
[14] Yang et al., Scientific Data 2023 (MedMNIST v2)
[15] Acevedo et al., Data in Brief 2020
[16] Hassini et al., Scientific Reports 2025
[17] Deb, Multi-Objective Optimization, Wiley 2001

Research gap and objective

The usual protocol

- one custom lightweight system vs. StyleGAN2-Ada [5]
- it can answer which of the two is better

What it cannot answer

- does the baseline belong on the frontier at all?
- a two-system design cannot test its own baseline [10], [16]

Testing the baseline requires a third lightweight system.

THE QUESTION THIS PAPER ANSWERS

Does StyleGAN2-Ada still sit on the empirical cost-quality curve at 64 × 64, once a third lightweight system enters the comparison?

[5] Karras et al., NeurIPS 2020 (StyleGAN2-Ada)
[10] Barrera et al., CMPB 2023 (SyntheticCellGAN)
[16] Hassini et al., Scientific Reports 2025

Systems compared

diffusion baselines excluded on operating-point grounds [13], [18]: multi-second inference per sample

OrGANiCells

- Vanilla DCGAN cascade [2] with R1 regularization
- One generator per class
- 2.65 M parameters, lightweight extreme

StyleGAN2-Ada

- NVlabs reference [5], class-conditional with ADA
- Pinned Python 3.8 / PyTorch 1.9.1
- 22.24 M parameters, the de-facto baseline

FastGAN-64

- Canonical lightweight reference [6], adapted here
- Small-branch plus reconstruction decoders
- 28.21 M parameters, our contribution

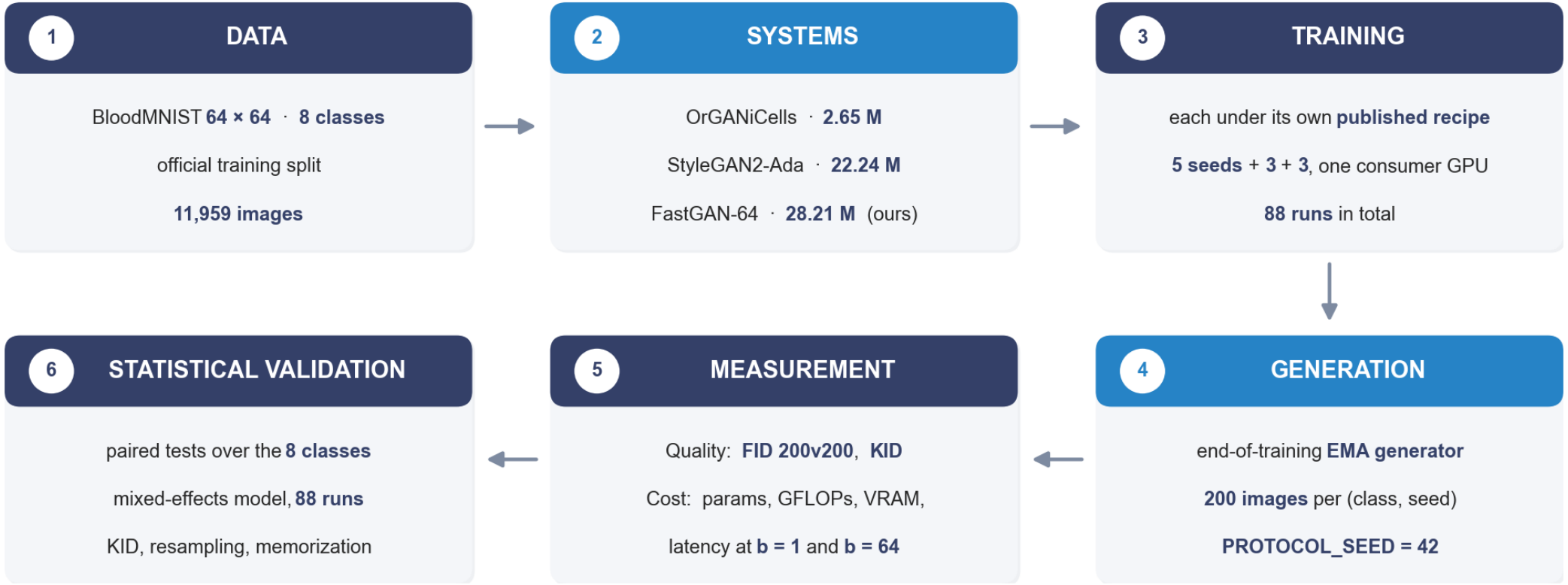
Architectural contribution: the reference FastGAN implementation hardcodes im_size ≥ 256 [6], and no public 64 × 64 instantiation existed. The dedicated Generator64 and Discriminator64 pair reuses the original primitives, so adding FastGAN as a baseline now costs about 62 GPU-hours.

Released: github.com/rfapo/lacci2026-lightweight-gan-tradeoff

[2] Radford et al., ICLR 2016 (DCGAN)
[5] Karras et al., NeurIPS 2020 (StyleGAN2-Ada)
[6] Liu et al., ICLR 2021 (FastGAN)
[13] Deltadahl et al., Nature Machine Intel. 2025
[18] Carreras Boada et al., ICCV-W CVAMD 2025 (CytoDiff)

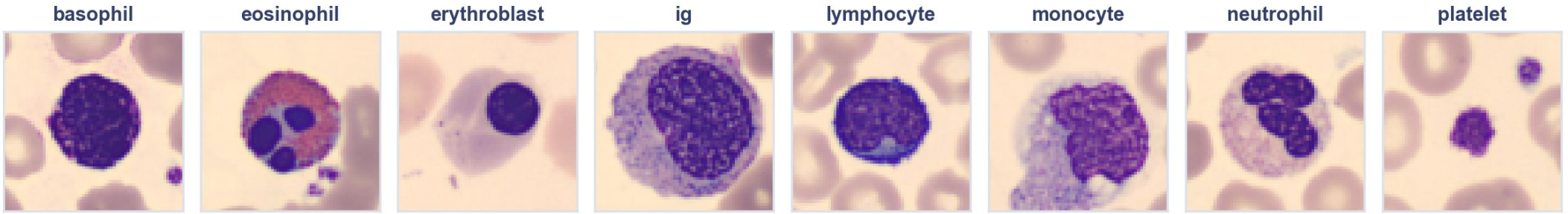
Experimental design

one pass through the whole study, from the dataset to the statistical validation; the next slide expands the evaluation protocol



Experimental protocol

- BloodMNIST 64 × 64, 8 classes, official training split (11,959 images) [14], [15]



One real sample from each of the eight classes, at the 64 × 64 resolution every system is trained and evaluated on.

- Per (class, seed): 200 generated images from the EMA generator vs. 200 real images; FID [19] on Inception-V3 features [20] computed with torch-fidelity [21], PROTOCOL_SEED = 42
- Identical protocol across all three systems, over 88 training runs in total (5 seeds for OrGANiCells, 3 for the others)

What this design gives up: each system trains under its own reference recipe, not a common budget (1 h, 23.5 h, 2.6 h per run), so the gap reflects architecture and recipe together.

[14] Yang et al., Scientific Data 2023 (MedMNIST v2)
[15] Acevedo et al., Data in Brief 2020
[19] Heusel et al., NeurIPS 2017 (FID)
[20] Szegedy et al., CVPR 2016 (Inception-V3)
[21] Obukhov et al., torch-fidelity, Zenodo 2020

Results: per-class FID 200v200

FID 200v200 [19]; mean ± std over seeds; lower is better; best per row in bold

Class	OrGANiCells	StyleGAN2-Ada	FastGAN-64
basophil	298.79 ± 24.55	54.57 ± 1.24	**48.93 ± 0.24**
eosinophil	219.24 ± 9.04	52.76 ± 11.03	**41.80 ± 1.15**
erythroblast	233.90 ± 13.16	73.94 ± 23.80	**47.23 ± 0.08**
ig	239.70 ± 14.25	49.67 ± 1.23	**48.42 ± 1.12**
lymphocyte	283.20 ± 22.53	66.71 ± 19.03	**42.73 ± 0.54**
monocyte	243.64 ± 12.20	57.03 ± 3.15	**50.87 ± 2.38**
neutrophil	229.83 ± 11.48	51.62 ± 6.94	**39.86 ± 0.50**
platelet	285.39 ± 9.16	70.11 ± 29.45	**42.47 ± 5.87**
Grand mean	**254.21 ± 31.82**	**59.55 ± 15.85**	**45.29 ± 4.32**

Lowest FID in

8 / 8 classes

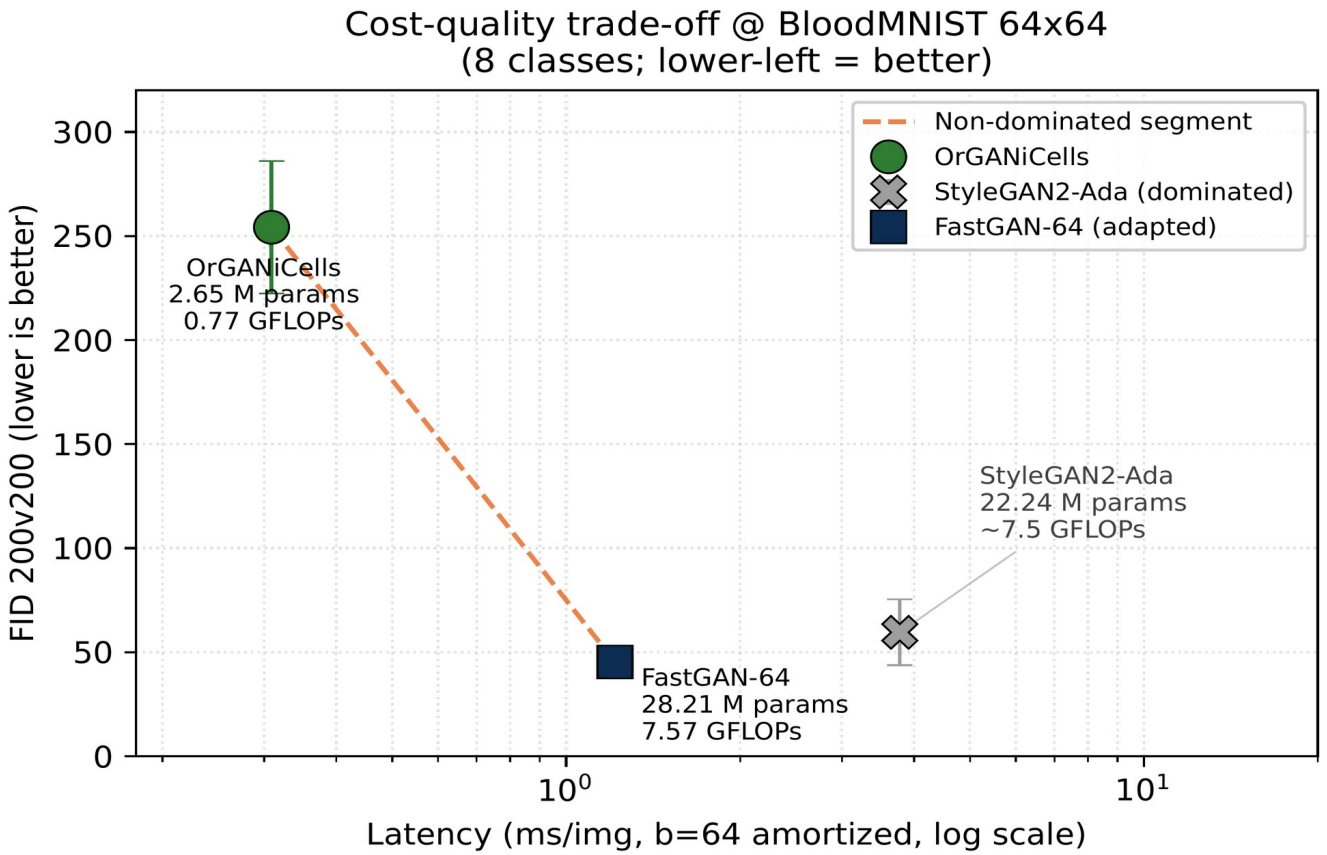
Lowest seed variance in

8 / 8 classes

StyleGAN2-Ada seed SD reaches 29.5 in platelet, 23.8 in erythroblast, 19.0 in lymphocyte

[19] Heusel et al., NeurIPS 2017 (FID)

Results: cost-quality trade-off



StyleGAN2-Ada is dominated

off the frontier on every deployment axis against FastGAN-64 (generator only):

Latency 3.1× faster at b = 64; 1.7× at b = 1

VRAM 1.57 GB down to 0.84 GB

FLOPs at parity (about 7.5 G)

Params the one axis kept, 22.2 vs. 28.2 M; the extra weights cost neither FLOPs nor VRAM

With three operating points this is a comparative trade-off analysis, not a strict Pareto front [17].

[17] Deb, Multi-Objective Optimization, Wiley 2001

Results: robustness checks

Could the gap be luck?

The same system wins in all eight classes, and the gap survives a model that accounts for class difficulty and for the scatter between training seeds.

+14.26 FID

Mixed-effects model, 88 runs

95% CI [3.41, 25.11]

Could it be the metric?

A second metric, built for small samples, and a fresh draw of real images give the same ordering.

8 / 8

Classes ordered alike by KID [22]

Kendall τ = 0.76

Could the winner be copying?

The winning generator is not reproducing the images it was trained on.

4.3%

Memorization audit

against 5% expected by construction

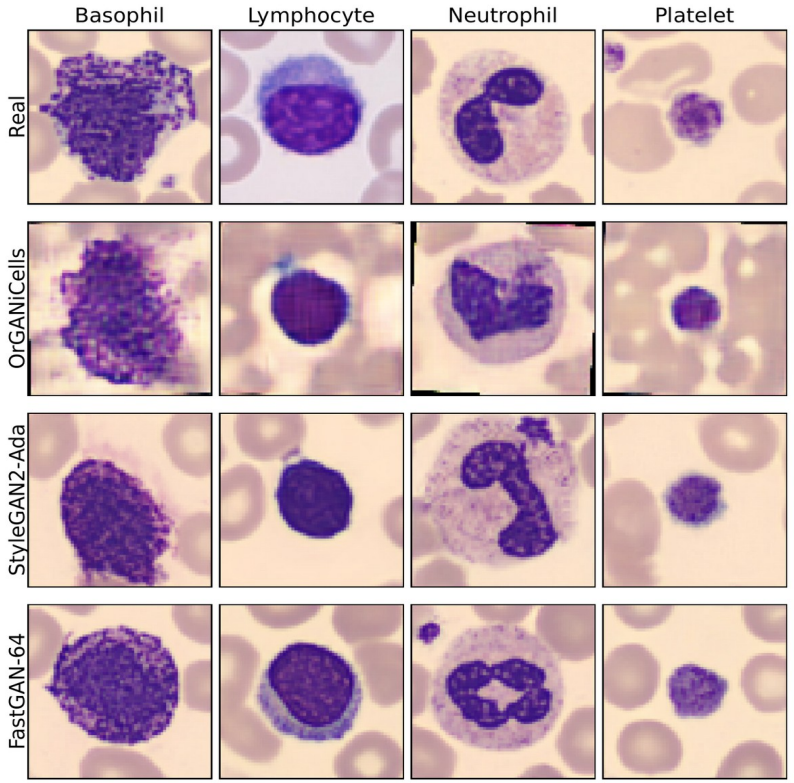
Three ways this result could have been false. None of them held. Full statistics in the paper [19], [22].

[19] Heusel et al., NeurIPS 2017 (FID)

[22] Binkowski et al., ICLR 2018 (KID)

Discussion

same image pools used for the quantitative evaluation



What the ordering means

OrGANiCells shows border and blur artifacts; both other systems preserve nuclear texture, so the visible ordering matches FID and KID

FID and KID run on Inception-V3 features trained on natural images [20], so no metric here establishes clinical realism, and downstream utility was not measured

Implications

Two-system comparisons against a heavy reference [10], [12], [16] can misattribute frontier position; such results are under-tested, not wrong

With the released patch, the third-system check costs about 62 GPU-hours

Scope: one dataset, one resolution, published training recipes, 3 to 5 seeds; the claim holds at this operating point, not for the lightweight regime in general

[10] Barrera et al., CMPB 2023 (SyntheticCellGAN)
[12] Eckardt et al., Leukemia 2022
[16] Hassini et al., Scientific Reports 2025
[20] Szegedy et al., CVPR 2016 (Inception-V3)

What we learned today

THE RESEARCH QUESTION

Does StyleGAN2-Ada still sit on the cost-quality curve at 64 × 64, once a third system enters?

No. It is displaced [5]. FastGAN-64 [6] wins in 8 of 8 classes, faster and lighter.

A two-system comparison could not have shown this.

THE PROVOCATION WE OPENED WITH

How light can a generative model be, and still support diagnosis?

Light enough for an 8 GB consumer GPU, at about a millisecond per image.

Best fidelity in the comparison; diagnostic utility is the next study.

Our hypothesis: when the deployable model is also the best one, access stops being a hardware problem.

[5] Karras et al., NeurIPS 2020 (StyleGAN2-Ada)

[6] Liu et al., ICLR 2021 (FastGAN)

Thanks!

Questions?

github.com/rfapo/lacci2026-lightweight-gan-tradeoff

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